Nishad **Singhi**

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Education

Present – Jul 2024 | **TU Darmstadt** — PhD in Computer Science — Advisors: Dr. Anna Rohrbach, Dr. Marcus Rohrbach

Mar 2024 – Nov 2020 | **University of Tübingen** — MSc in Neural Information Processing — Advisors: Dr. Zeynep Akata, Dr. Wieland Brendel

Apr 2020 – Aug 2016 | **Indian Institute of Technology (IIT) Delhi** — BTech in Electrical Engineering — Advisor: Dr. Sumeet Agarwal

Select Experience

Sep 2026 – Apr 2026 | **Apple Machine Learning Research** — **Cupertino, CA**
Research Intern | Advisors: Dr. Alex Toshev, Dr. Andrew Szot
Worked on self-improving data generation for tool-use agents, where generator agents diagnose a learner agent's shortcomings and adaptively create new training tasks to close them.

May 2026 – Jun 2025 | **OpenThoughts-Agent**
Open-source Contributor | Advisor: Dr. Ludwig Schmidt
Contributed to data curation and evaluation for Terminal Agents.

Research Interests

LLM Agents, Multimodal AI, Reasoning, Embodied AI

Select Publications

* = Equal Contribution

Reasoning with Image Generation — **COLM 2026**
N. Singhi*, H.G. Rodriguez*, A. Arora, M. Rohrbach, A. Rohrbach. [Paper]
We propose using image generation models to create visual artifacts that improve multimodal reasoning.

OpenThoughts-Agent: Data Recipes for Agentic Models — **arXiv 2026**
N. Raoof, ..., N. Singhi, ..., L. Schmidt. [Paper] [Website]
An open data-curation pipeline for training agentic language models.

Think Twice, Act Once: Verifier-Guided Action Selection for Embodied Agents — **CVPR 2026 (Findings)**
N. Singhi, C. Bialas, S. Jauhri, V. Prasad, G. Chalvatzaki, M. Rohrbach, A. Rohrbach. [Paper]
We improve OOD generalization in multimodal embodied agents by training generative verifiers to discard incorrect actions.

When To Solve, When To Verify: Compute-Optimal Problem Solving and Generative Verification for LLM Reasoning — **COLM 2025**
N. Singhi*, H. Bansal*, A. Hosseini*, A. Grover, K.W. Chang, M. Rohrbach, A. Rohrbach. [Paper] [Code]
We study how to best allocate test-time compute, finding that majority voting beats gen. verifier except at high inference costs.

Improving Intervention Efficacy via Concept Realignment in Concept Bottleneck Models — **ECCV 2024**
N. Singhi, J. Kim, K. Roth, Z. Akata. [Paper] [Talk] [Code]
We introduce a lightweight module that makes CBMs more efficient by minimizing the number of interventions required.

CleanCLIP: Mitigating Data Poisoning Attacks in Multimodal Contrastive Learning — **ICCV 2023 (Oral)**
H. Bansal*, N. Singhi*, Y. Yang, F. Yin, A. Grover, K.W. Chang. [Paper] [Talk] [Code]
Best Paper Award at the Reliable and Trustworthy ML Workshop at ICLR 2023
Proposed a technique to remove backdoor attacks from CLIP by independently finetuning vision and language features.

Toward a normative theory of (self-)management by goal-setting — **CogSci 2023 (Oral)**
N. Singhi, F. Mohnert, B. Prystawski, F. Lieder. [Paper] [Talk]
Diversity and Inclusion Award (10 recipients worldwide)
Built a tool to automatically decompose challenging problems into easier subgoals for humans, leading to 3x reduction in effort.

Talks

Building Capable Multimodal Agents via Test-Time Reasoning, UC Berkeley — Sep 2026
Toward a normative theory of (self-)management by goal-setting, CogSci [Talk] — Jul 2023
CleanCLIP: Mitigating Data Poisoning Attacks in Multimodal Contrastive Learning, RTML @ ICLR [Talk] — May 2023